

TAHIA ANAN DHIRA

tahiaanand@gmail.com | dhiraanan@du.ac.bd | tahiaanandhira.com | github.com/TAD600

EDUCATION

MSc in Applied Social Data Science (Distinction) — London School of Economics and Political Science	2023
<i>Coursework: Social Network Analysis; Quantitative Text Analysis; Applied Machine Learning; Managing and Visualizing Data; Research Design</i>	
Master of Social Science (MSS) in Economics (CGPA 3.88/4.00) — University of Dhaka	2017
Bachelor of Social Science (BSS) in Economics (CGPA 3.94/4.00) — University of Dhaka	2016

AWARDS

Commonwealth Master's Scholarship	2022
Gold Medals for highest CGPA, BSS & MSS Economics, University of Dhaka	2016, 2017

CURRENT RESEARCH MANUSCRIPTS

Gender Stereotypes in Social Media Audience Discourse: Evidence from Bangladeshi Artists

(Working paper, 2026) | Corresponding author

Sentiment, keyness, and topic modeling of YouTube comments show that audience gender stereotypes vary systematically across artistic professions.

- Built scalable Python pipelines using YouTube Data API v3 to collect and process multilingual comment corpus
- Developed an annotation workflow in which the Gemini API generated preliminary art-category and gender classifications from video titles, followed by manual coding, correction, and validation of the full video corpus

Bilateral Convergence across the World: Does International Migration Matter?

(Working paper, 2026) | coauthors: M. Moniruzzaman, S. N. Wadood

A country's core or peripheral position in the global migration network, not migration volume alone, shapes whether migration narrows or widens development gaps.

- Built a 23-year dynamic panel of directed, weighted bilateral migration networks
- Contributed to network model development, core-periphery detection, and manuscript preparation

Product Reproducibility and International Trade Patterns: Evidence from Cultural Goods

(Under review, 2026) | Corresponding author

More reproducible products are less sensitive to trade friction and follow gravity predictions more closely.

- Constructed computational pipelines to harmonize and process bilateral trade data across 190 countries

Rethinking the Income–Emissions Relationship: Evidence from Trade-Embodied CO₂ Emissions Networks

(Submitted, 2026) | Corresponding author

Income turning points for emissions shift with a country's network centrality in the trade-embodied carbon network.

- Built reproducible computational workflows integrating large-scale international trade-embodied emissions and macroeconomic datasets
- Constructed weighted global trade-embodied-emissions networks, computed network metrics and country positions

PUBLICATIONS

Huque, R., Abdullah, S. M., **Dhira, T. A.**, Siddiqua, S., Barua, D., Abdullah, Md., Sabur, M. A., Begum, N., Dogar, O., Amin, Md. N., & Mahmud, Mohd. S. H. (2025). Unveiling health insurance satisfaction: Exploring key determinants and bottlenecks in Bangladesh. *SSM - Health Systems*, 4, 100058.
<https://doi.org/10.1016/J.SSMHS.2025.100058>

Rahman, M. A., **Dhira, T. A.**, Sarker, A. R., & Mehareen, J. (2022). Validity and reliability of the Patient Health Questionnaire scale (PHQ-9) among university students of Bangladesh. *PLoS ONE*, 17(6).
<https://doi.org/10.1371/journal.pone.0269634>

Dhira, T. A., Rahman, M. A., Sarker, A. R., & Mehareen, J. (2021). Validity and reliability of the Generalized Anxiety Disorder-7 (GAD-7) among university students of Bangladesh. *PLoS ONE*, 16(12).

<https://doi.org/10.1371/journal.pone.0261590>

Mehareen, J., Rahman, M. A., **Dhira, T. A.**, & Sarker, A. R. (2021). Prevalence and socio-demographic correlates of depression, anxiety, and co-morbidity during COVID-19: A cross-sectional study among public and private university students of Bangladesh. *Journal of Affective Disorders Reports*, 5.

<https://doi.org/10.1016/j.jadr.2021.100179>

EARLIER RESEARCH EXPERIENCE

Capstone Project, MSc ASDS, LSE

Friendship Network Formation: Pro-sociality and Personality Type (2023)

- Designed and implemented a behavioural classroom experiment on homophily and friendship network formation
- Estimated Exponential Random Graph Models (ERGMs) to identify structural and attribute-based drivers of network formation

Coursework project, ST445, LSE

Topic-Based Content and Sentiment Analysis of COVID-19 Vaccine Discourse on Reddit (2022–2023)

- Visualized a subreddit-level network in which two subreddits are linked by shared frequent commenters among the 40 most active authors, identifying the most central communities and informing the choice to scope subsequent analysis to the top 20 subreddits
- Contributed to report writing, documentation, and referencing on a project led by a classmate, who collected, processed, and modeled the underlying multi-million-comment corpus

Mental Health of University Students Before and After Online Education (UGC-funded) (2020–2021)

- Cleaned survey data and validated research instruments; contributed to preparation of project reports

Willingness to Pay for Treating Mental vs. Physical Illness (BMRC-funded) (2020–2021)

- Cleaned survey data and contributed to instrument validation and report preparation for a study on adults

ACADEMIC APPOINTMENTS

Assistant Professor, Department of Economics, University of Dhaka

October 2025 – present

Lecturer, Department of Economics, University of Dhaka

November 2018 – September 2025

- Taught 10 undergraduate economics courses; supervised undergraduate dissertations and research projects
- Trained undergraduate and graduate students in STATA; developed computational course materials in Python and STATA

Lecturer, Department of Economics, East West University

March 2018 – October 2018

COMPUTATIONAL & QUANTITATIVE SKILLS

- **Econometrics:** panel-data and time-series methods; structural gravity/PPML estimation; factor analysis, principal component analysis, structural equation modelling
- **Network analysis:** ERGM, STERGM, dynamic and weighted networks, centrality, community detection, core-periphery analysis, block modelling, cluster analysis
- **Softwares and packages:** STATA · Python (pandas, NumPy, NetworkX, Matplotlib, scikit-learn) · R (igraph, statnet, ergm, tergm, stm, networkDynamic, quanteda)
- **Data collection & construction:** API-based and web-scraped data collection; Qualtrics; harmonizing and constructing large-scale, multi-source panel datasets from trade, migration, survey, and administrative sources
- **Machine learning & NLP:** applied machine learning; sentiment analysis (VADER), clustering (structural topic modelling, LDA), applied machine learning; word embeddings, LDA, ideological scaling (Wordscores, Wordfish), and transformer-based text classification (BERT)
- **Research workflow:** Git/GitHub; Gemini API; reproducible pipelines; data visualization; LaTeX

LANGUAGES

Advanced fluency in English and Bangla